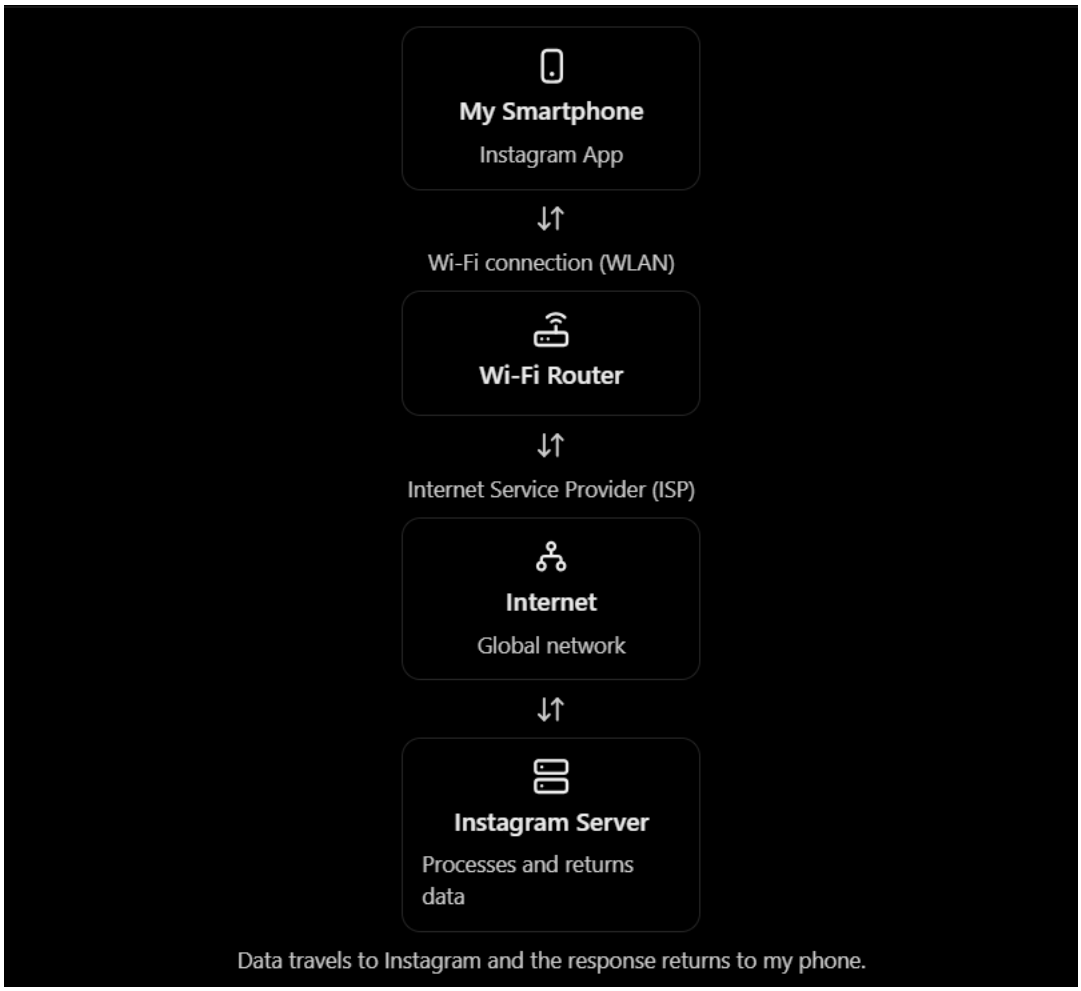
Network Fundamentals

1. Diagram: How My Phone Connects to Instagram



Explanation: When I open Instagram, my smartphone sends a request through the Wi-Fi router and ISP over the Internet. Instagram's server processes the request and sends the required data back to my phone.

2. Five Network Devices I Use or See Daily

Device	Type of Network	Use
Smartphone	WLAN / WAN	Connects to Wi-Fi or mobile data.
Laptop	LAN / WLAN	Connects to the Internet through Wi-Fi or Ethernet.
Wi-Fi Router	LAN	Connects multiple devices within a local network.
Smart TV	LAN / WLAN	Streams videos using an Internet connection.
Smartwatch	PAN	Connects to a smartphone using Bluetooth.

3. Difference Between Internet and Intranet

Internet	**Intranet**
A global network accessible to the public.	A private network accessible to authorized users.
Used to access websites such as Google and Instagram.	Used to access an organization's private resources.
Requires an Internet connection for online services.	May work within an organization's network without public Internet access.

Example from my college:

- Internet: I use the Internet to search for study materials on Google or watch educational videos on YouTube.
- Intranet: My college may provide a private student portal accessible to authorized students and staff for internal notices, attendance, or academic records.

4. How Data Travels When I Search on Instagram

1. I open Instagram on my smartphone and type a name into the search bar.
2. When I press the Search button, my phone sends a request through Wi-Fi and the router to my Internet Service Provider (ISP).
3. The request travels over the Internet to Instagram's server.
4. The server processes the request, finds relevant accounts or content, and sends the results back through the Internet.
5. My smartphone receives the response and displays the search results in the Instagram app.